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From Data to Decisions

MODULE 6

SAMPLING DISTRIBUTION & CENTRAL LIMIT THEOREM

From Samples to **Reliable Decisions**

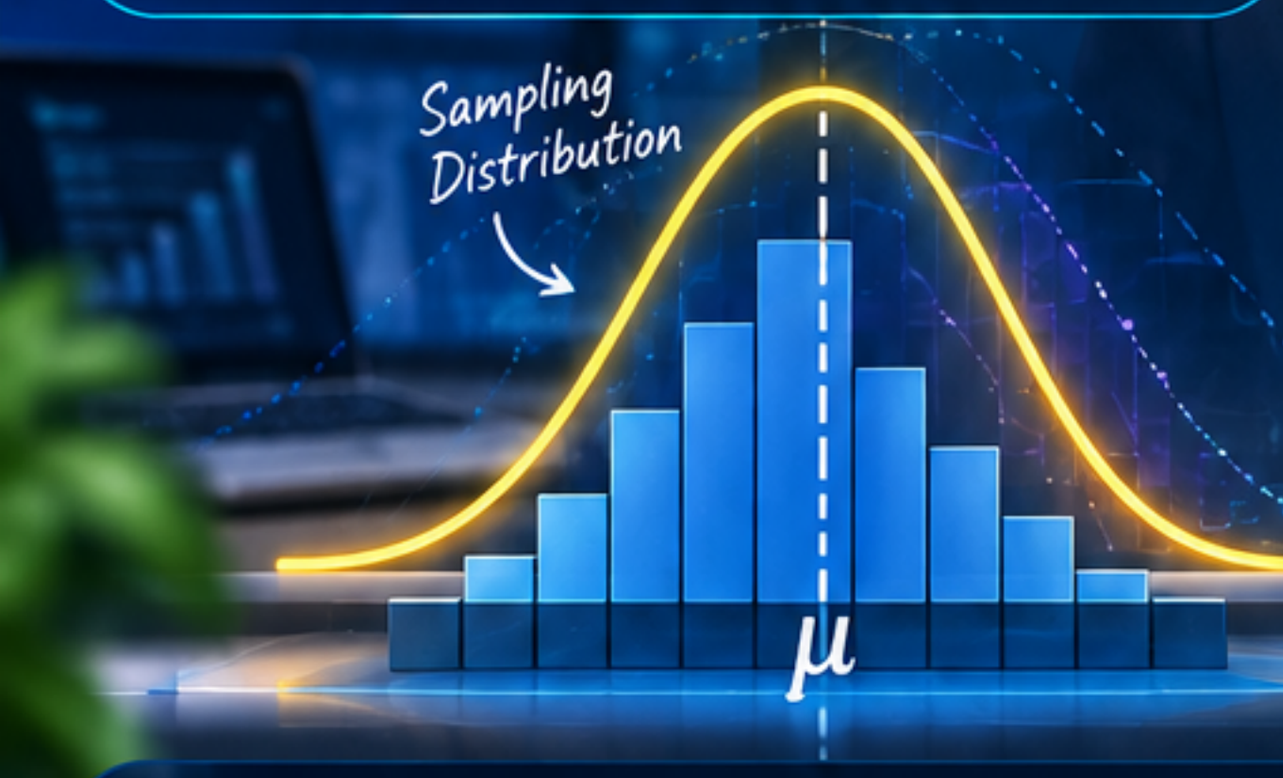
TOPICS COVERED

1. Sampling Distribution of the Mean
2. Sampling Distribution of Proportions
3. Central Limit Theorem (CLT)
4. Conditions for CLT
5. Standard Error
6. Effect of Sample Size
7. Point Estimate vs Interval Estimate
8. Bias and Unbiased Estimators

Statistics for Real World Insights

SAMPLES BRING CLARITY TO DATA





Central Limit Theorem

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$



Concepts Made Simple | Real-World Examples | Industry Use Cases | Interview Ready

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DATA ANALYTICS | EXCEL | STATISTICS | REAL BUSINESS SKILLS

1 Sampling Distribution of the Mean

Understanding how sample means behave and why they are important in real-world data analysis.

What is Sampling Distribution?

The sampling distribution is the **probability distribution of a statistic** (like sample mean) calculated from many possible samples of the same size taken from a population.

Sampling Distribution of the Mean

The sampling distribution of the mean ($\overline{X}$) is the **distribution of all possible sample means** calculated **from repeated random samples of size n** drawn from the same population.

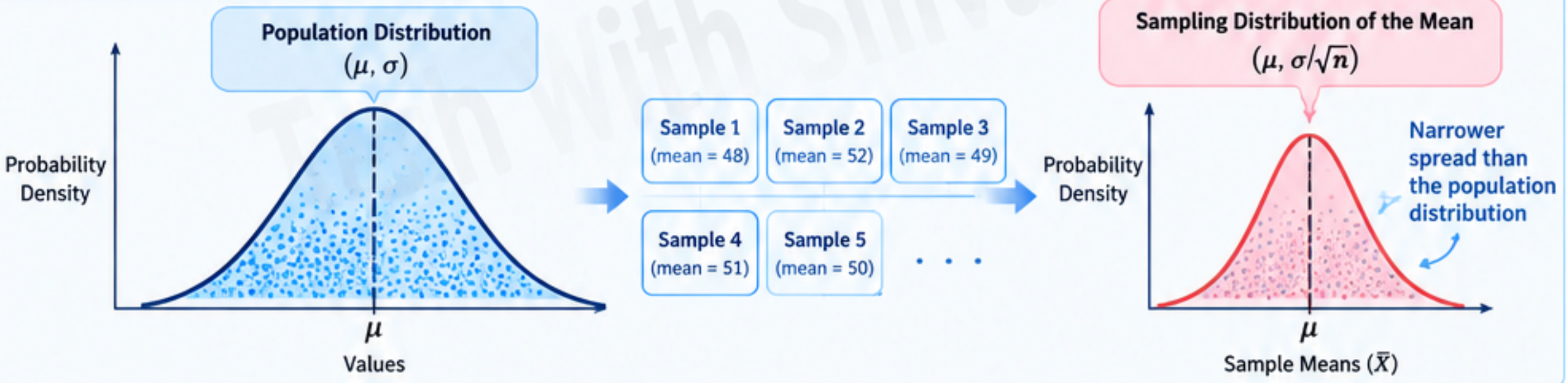
Why is it Important?

1. Helps in estimation of population parameters.
2. Forms the basis for confidence intervals.
3. Critical for hypothesis testing.
4. Used widely in business decision making.
5. Works even when the population distribution is unknown (due to CLT).

Key Points to Remember

- Every sample gives a different sample mean.
- If we take many samples, the sample means form a distribution.
- This distribution is called the sampling distribution of the mean.
- The mean of the sampling distribution equals the population mean (μ).
- The spread of the sampling distribution is smaller than the population distribution.

Visual Representation



Mathematical Properties

Mean of Sampling Distribution

$$E(\overline{X}) = \mu$$

The mean of the sampling distribution of the mean is equal to the population mean.

Standard Deviation (Standard Error)

$$\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}$$

The standard deviation of the sampling distribution of the mean is called the **standard error** (SE).

Interpretation

- Sample means are centered around the population mean (μ).
- As sample size (n) increases, the spread (standard error) decreases.
- The distribution of sample means is more normally distributed than the population distribution (due to CLT).

Example: Customer Spending

Scenario

A company wants to estimate the average monthly spending of its customers. The **population** mean is ₹5,000 and the population **standard** deviation is ₹1,200. If we take **random samples** of 36 customers, what is the sampling distribution of the mean?

Solution

Given:

$\mu = 5{,}000$
$\sigma = 1{,}200$
$n = 36$

Sampling Distribution:

$$E(\overline{X}) = \mu = 5{,}000$$

$$\sigma_{\overline{X}} = \frac{1{,}200}{\sqrt{36}} = \frac{1{,}200}{6} = 200$$

Interpretation

If we repeatedly take samples of **36** customers, the sample means will be distributed around ₹5,000 with a standard error of ₹200.

Business Use Case

Helps the company understand how much the sample average can vary from the true population average, which is essential for confidence intervals and making reliable business decisions based on sample data.

2 Sampling Distribution of Proportions

Understanding how sample proportions behave and how they help in real-world decision making.

What is Sampling Distribution of Proportions?

The sampling distribution of a proportion ($\hat{p}$) is the **distribution of all possible sample proportions** calculated from many random samples of size n taken from the same population.

Population Proportion vs Sample Proportion

Population Proportion (p)

The true proportion in the entire population.

$$p = \frac{X}{N}$$

X = number of successes
N = population size

Sample Proportion ($\hat{p}$)

The proportion calculated from a sample.

$$\hat{p} = \frac{x}{n}$$

x = number of successes in the sample
n = sample size

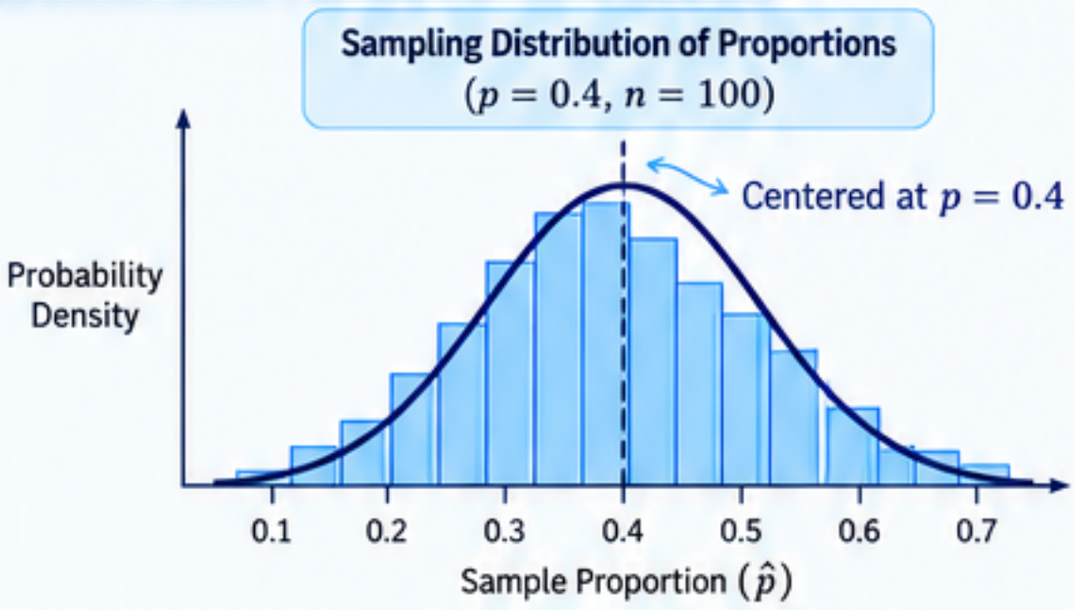
Why is it Important?

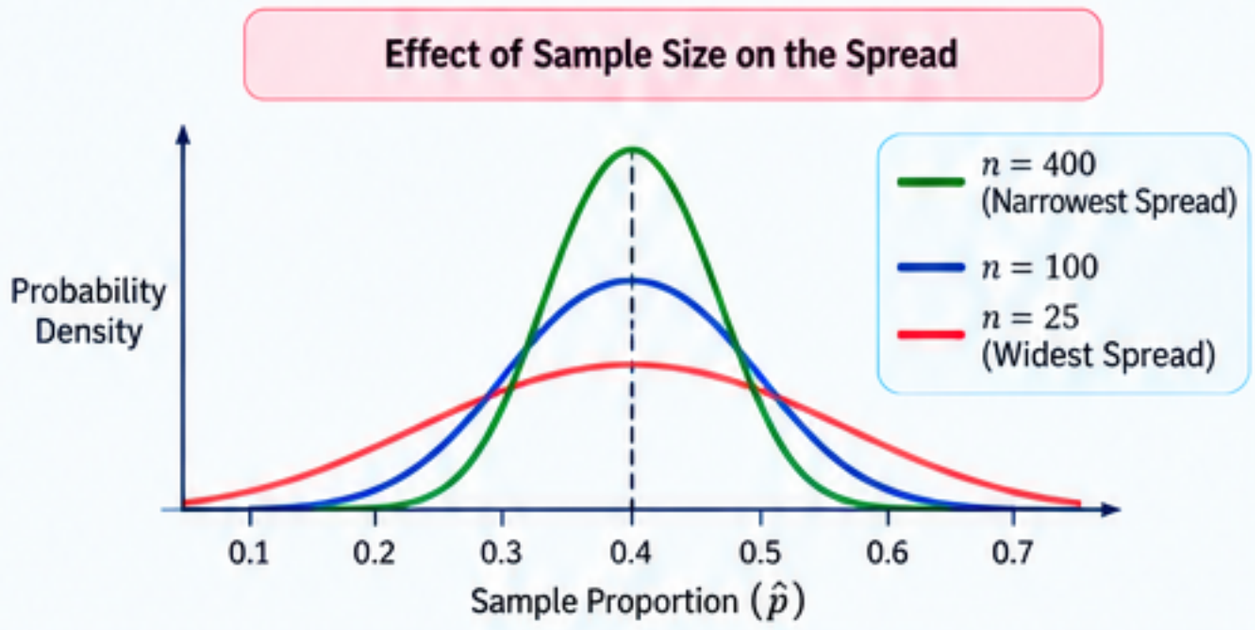
1. Used to estimate population proportions.
2. Forms the basis for confidence intervals.
3. Critical for hypothesis testing (e.g., A/B testing conversion rates).
4. Widely used in business, marketing and surveys.
5. Helps in decision making when only sample data is available.

Key Points to Remember

- Each sample gives a different sample proportion.
- The sample proportions form a distribution.
- The mean of the sampling distribution of a proportion equals the population proportion (p).
- The spread of the distribution is measured by the standard error.
- For large enough sample sizes, the distribution of sample proportions is approximately normal (even if the population distribution is not).

Visual Representation





Mathematical Properties

Mean of Sampling Distribution of Proportion

$$E(\hat{p}) = p$$

The mean of the sample proportion is equal to the population proportion.

Standard Error (SE)

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

The standard deviation of the sampling distribution of the proportion is called the **standard error**.

Conditions for Approximate Normality

1. n is sufficiently large
2. $np \geq 10$
3. $n(1-p) \geq 10$

When these conditions are satisfied, the sampling distribution of $\hat{p}$ is approximately normal.

Example: E-commerce Conversion Rate

Scenario

An e-commerce website has a true conversion rate of 8% ($p = 0.08$).
A random sample of **200** visitors is selected. What is the sampling distribution of the sample proportion?

Solution

Given:

$p = 0.08$
$n = 200$

Mean: $E(\hat{p}) = p = 0.08$

Standard Error:

$$\sigma_{\hat{p}} = \sqrt{\frac{0.08(1-0.08)}{200}}$$

$$= \sqrt{\frac{0.0736}{200}} = 0.0192$$

Interpretation

- On average, the sample proportion will be 0.08 (8%).
- Most sample proportions will lie close to 0.08 with a standard error of 0.0192.
- This helps us understand the variation we may expect in sample conversion rates across different samples of size 200.

3 Central Limit Theorem (CLT)

Samples lead to Normal Distribution

The most important theorem in statistics that connects samples to the normal distribution.

What is Central Limit Theorem (CLT)?

The Central Limit Theorem states that the **distribution of the sample mean** becomes **approximately normal** as the sample size increases, **regardless of the shape of the population distribution** (provided the population has a finite mean and standard deviation).

Why is it Important?

1. Allows us to use normal distribution for inference even when the population is not normal.
2. Forms the basis for confidence intervals and hypothesis testing.
3. Widely used in business, analytics and data science.
4. Helps in making reliable decisions from sample data.
5. Works well with reasonably large sample sizes.

Key Conditions for CLT

1. Random sampling (observations are independent).
2. Sample size should be sufficiently large.
3. Population must have a finite mean (μ) and finite standard deviation (σ).
4. CLT works for any population shape (normal, skewed, uniform, etc.).

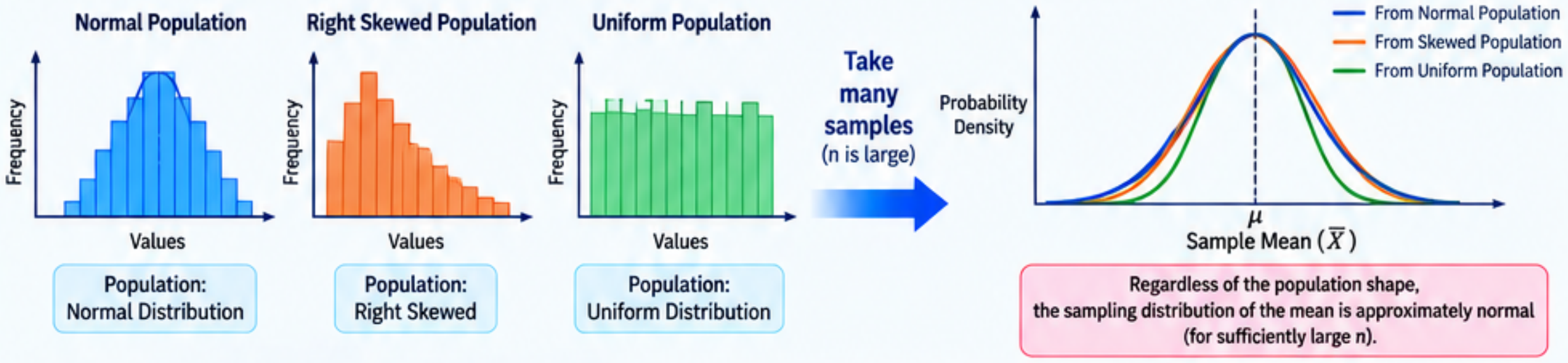
Mathematical Statement

If the population has mean μ and standard deviation σ, then for a sufficiently large sample size n, the sampling distribution of the mean ($\overline{X}$) is approximately normal with:

$$\overline{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

μ = population mean
σ = population standard deviation
n = sample size
$\frac{\sigma}{\sqrt{n}}$ = standard error

Visual Explanation of CLT



Example 1: Right Skewed Population

Scenario

A company's customer purchase amount is right skewed. The population mean is ₹1,000 and the population standard deviation is ₹600. If we take random samples of size $n = 100$, what is the distribution of the sample mean?

Solution

- Population is right skewed, but CLT applies (since $n = 100$ is large).
- Sampling distribution of the mean is approximately normal.
- Mean of $\overline{X} = \mu = 1{,}000$
- Standard error $= \frac{\sigma}{\sqrt{n}} = \frac{600}{\sqrt{100}} = \frac{600}{10} = 60$

$$\overline{X} \sim N(1{,}000, 60)$$

Interpretation

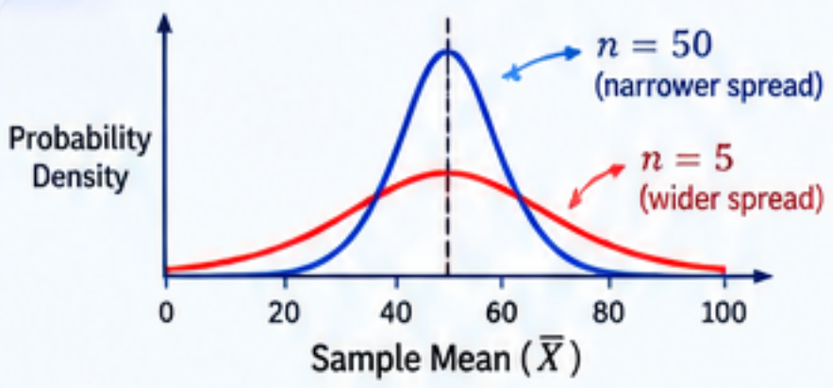
- Even though the original data is skewed, the sample means follow a normal distribution.
- Most sample means will lie close to ₹1,000 with a standard error of ₹60.
- This allows us to use normal distribution for confidence intervals and hypothesis testing.

Example 2: Small vs Large Sample Size

Scenario

A population is uniform (all values between 0 and 100). Compare the sampling distribution of the mean for $n = 5$ and $n = 50$.

Visual Comparison



Key Takeaway

- Both distributions are approximately normal due to CLT.
- As sample size increases, the sampling distribution becomes narrower (smaller standard error).
- The mean remains the same (μ), but the spread decreases.

4 Conditions for CLT

Key conditions under which the Central Limit Theorem holds and why they matter.

When these conditions are satisfied, we can use the normal distribution for inference even if the population is not normal.

What are the Conditions for CLT?

The Central Limit Theorem works under **certain conditions**. When these conditions are satisfied, the sampling distribution of the mean (or proportion) becomes **approximately normal**, even if the population **distribution is unknown or not normal**.

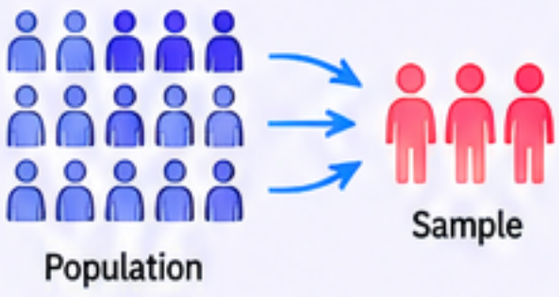
Why are these conditions important?

1. Ensure that the sample means are independent.
2. Reduce the effect of extreme values.
3. Allow the sampling distribution to stabilize and become normal.
4. Give reliable results for confidence intervals and hypothesis testing.
5. Help us decide the appropriate sample size.

Main Conditions for CLT (for Sample Mean)

1 Random Sampling

The sample must be selected randomly from the population so that each observation is **independent**.



2 Independence

Observations in the sample should be independent of each other.

- One observation should not influence another.
- This is usually satisfied when sampling without replacement or when the population is very large ($> 10n$).

3 Sufficiently Large Sample Size

The sample size should be large enough. A common rule of thumb is:

$$n \geq 30$$

For many real-world cases, $n \geq 30$ is sufficient even if the population is not normal.

4 Finite Mean and Standard Deviation

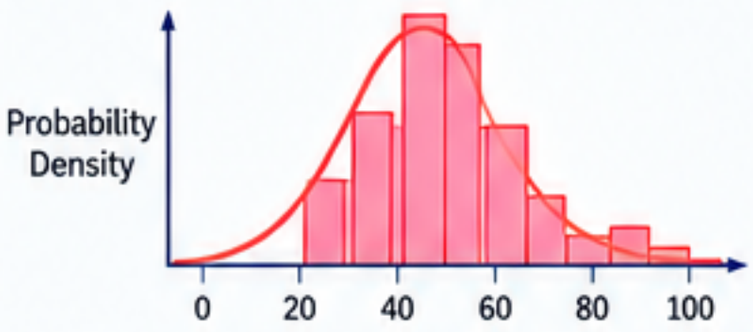
The population should have a finite mean (μ) and finite standard deviation (σ).

$$\mu < \infty \quad \text{and} \quad \sigma < \infty$$

- CLT does not hold if the population has an infinite mean or infinite variance (e.g., some extreme heavy-tailed distributions).

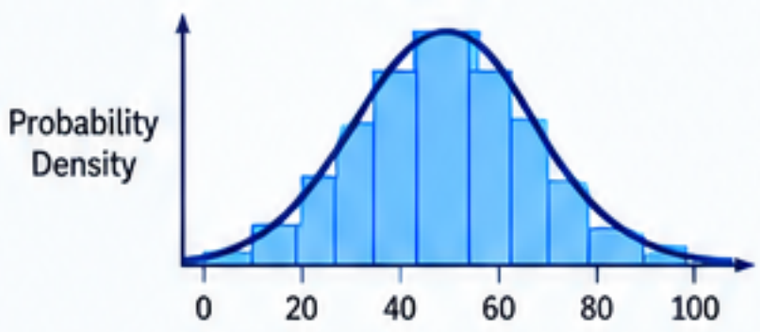
Visual Illustration of Conditions

1 Small Sample ($n = 5$)



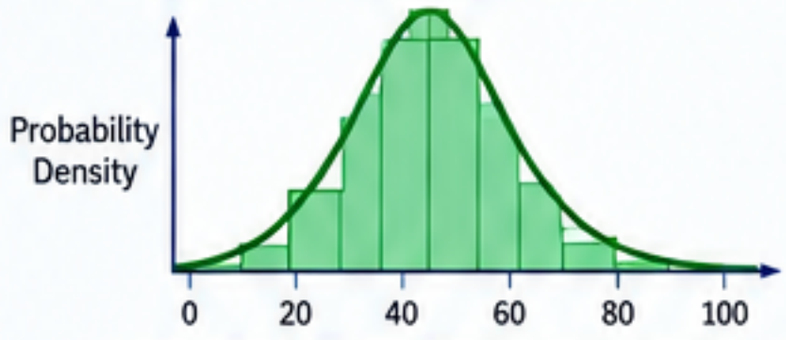
For small sample sizes, the sampling distribution may not look normal.

2 Moderate Sample ($n = 30$)



As sample size increases, the distribution becomes more symmetric.

3 Large Sample ($n = 100$)



For large sample sizes, the sampling distribution is approximately normal, even if the population is not normal.

Example: Population is Not Normal (Skewed Data)

Scenario

A company analyzes the time taken to resolve customer complaints. The population distribution is right skewed with mean 50 minutes and standard deviation 20 minutes. We take random samples of size $n = 5$, 30 and 100. What can we say about the sampling distribution of the mean?

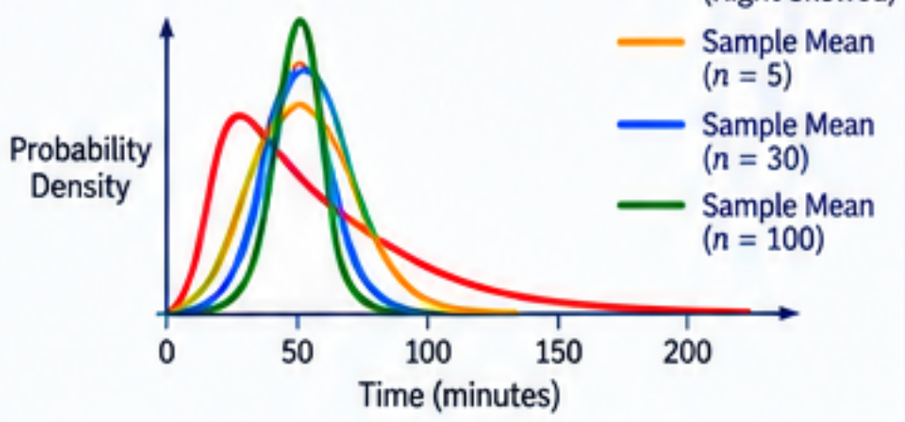
Solution

Sample Size (n)	Sampling Distribution of $\bar{X}$
$n = 5$	Still skewed, not close to normal.
$n = 30$	More symmetric, closer to normal.
$n = 100$	Approximately normal (due to CLT).

Mean of the sampling distribution $= \mu = 50$

$$\text{Standard error} = \frac{\sigma}{\sqrt{n}} = \frac{20}{\sqrt{n}}$$

Visual Comparison



Even though the original population is right skewed, the sampling distribution of the mean becomes approximately normal as the sample size increases.

5 Standard Error

Understanding the standard error and how it measures the variability of a sample statistic.

Standard error tells us how much the sample statistic can vary across different samples.

What is Standard Error?

The **standard error (SE)** is the **standard deviation of the sampling distribution of** a statistic. It measures how much the sample statistic (like the sample mean or sample proportion) is expected to vary from sample to sample.

Why is it Important?

1. Shows the precision of our sample estimate.
2. Helps in constructing confidence intervals.
3. Used in hypothesis testing.
4. Indicates the reliability of the sample statistic.
5. Decreases as sample size increases (more data = more precise estimates).

Standard Error of the Mean

The standard error of the sample mean ($\overline{X}$) is:

$$SE(\overline{X}) = \frac{\sigma}{\sqrt{n}}$$

σ = population standard deviation
n = sample size

- It tells us how much the sample mean is expected to vary across different samples.
- A smaller standard error indicates a more precise estimate of the population mean.

Standard Error of the Proportion

The standard error of the sample proportion ($\hat{p}$) is:

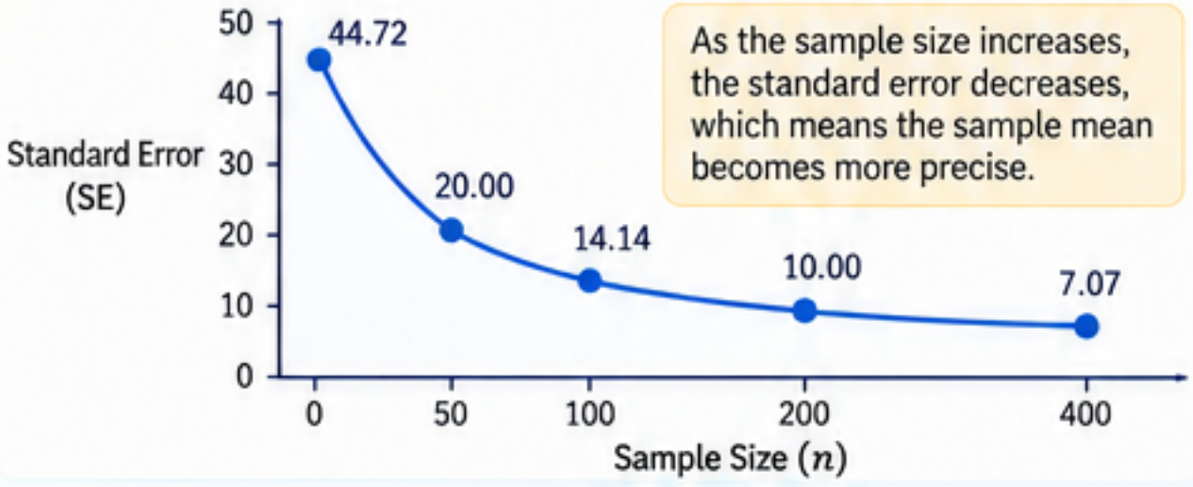
$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

p = population proportion
n = sample size

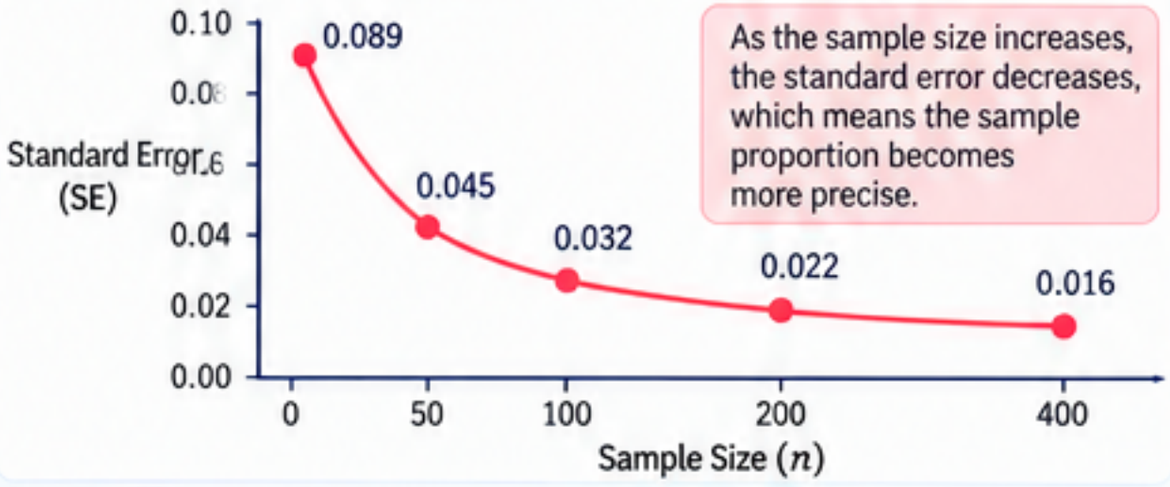
- It tells us how much the sample proportion is expected to vary across different samples.
- A smaller standard error indicates a more precise estimate of the population proportion.

Visualizing the Effect of Sample Size on Standard Error

Standard Error of the Mean



Standard Error of the Proportion



Example 1: Standard Error of the Mean (Delivery Time)

Scenario

The population standard deviation of delivery time is 15 minutes. A random sample of 100 orders is selected. What is the standard error of the sample mean?

Solution

Given:

$\sigma = 15$
$n = 100$

$$SE(\overline{X}) = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{100}} = \frac{15}{10} = 1.5 \text{ minutes}$$

Interpretation

- The sample mean delivery time is expected to vary by about 1.5 minutes across different samples of 100 orders.
- If we take multiple samples of 100 orders, most sample means will be within a small range around the population mean.

Example 2: Standard Error of the Proportion (Conversion Rate)

Scenario

A website has a true conversion rate of 10% ($p = 0.10$). A random sample of 400 visitors is selected. What is the standard error of the sample proportion?

Solution

Given:

$p = 0.10$
$n = 400$

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.10(1-0.10)}{400}} = \sqrt{\frac{0.09}{400}} = \sqrt{0.000225} = 0.015$$

Interpretation

- The sample proportion (conversion rate) is expected to vary by about 0.015 (1.5%) across different samples of 400 visitors.
- This means most sample proportions will lie close to 10% with a standard error of 1.5%.

6 Effect of Sample Size

How sample size (n) affects the sampling distribution, standard error and accuracy of estimates.

Larger sample size leads to a more precise and stable estimate.

What is the Effect of Sample Size?

The **sample size** (n) directly affects the **spread** of the sampling distribution. As the **sample size** increases, the **standard error decreases**, which means the sample statistic (like mean or proportion) becomes **more precise** and **less variable**.

Key Takeaways

1. Larger sample size → smaller standard error.
2. Sampling distribution becomes narrower and more concentrated around the true parameter.
3. Estimates become more reliable and precise.
4. Helps in confidence intervals and hypothesis testing.
5. After a certain point, increasing n gives smaller improvements.

Mathematical Relationship

Standard Error of the Mean

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

As n increases, SE decreases at a rate of $\frac{1}{\sqrt{n}}$.

Standard Error of the Proportion

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

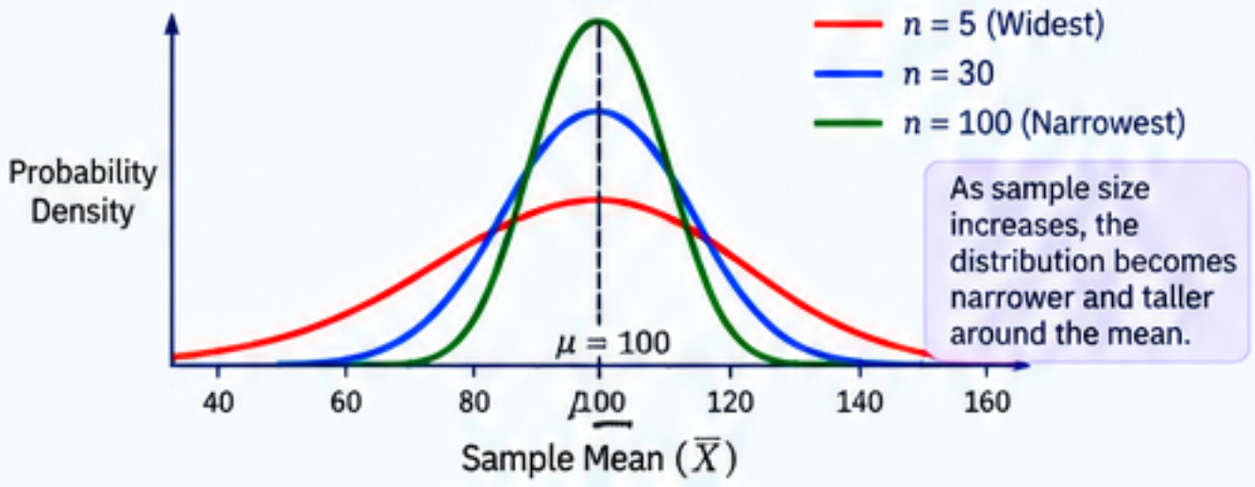
As n increases, SE decreases at a rate of $\frac{1}{\sqrt{n}}$.

Intuition Behind the Effect

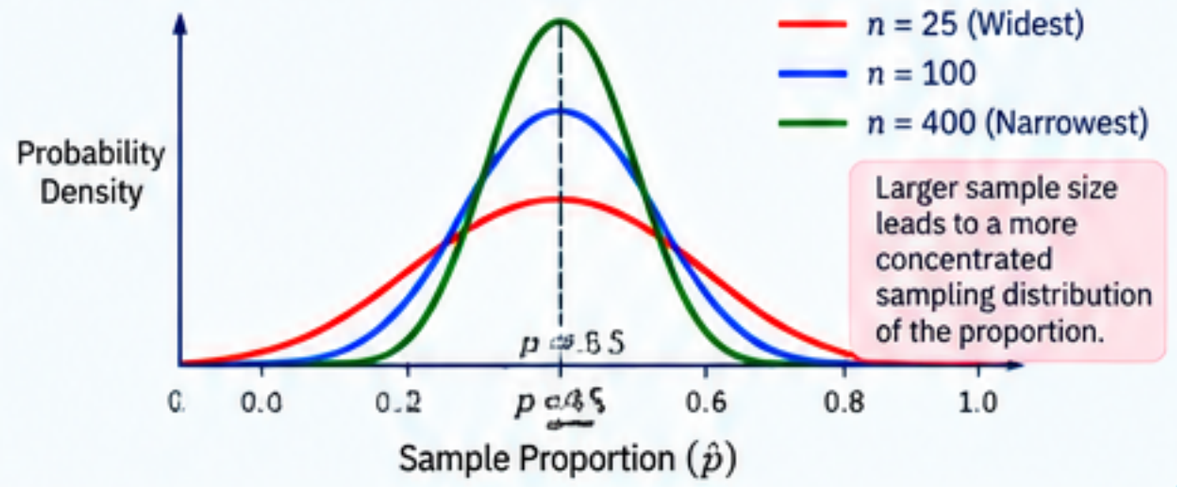
- A larger sample size reduces the impact of random fluctuations in the data.
- With more observations, the sample statistic becomes closer to the true population parameter.
- The variation between different samples decreases.
- This results in a narrower sampling distribution.

Visualizing the Effect of Sample Size on Sampling Distribution

Sampling Distribution of the Mean (Different Sample Sizes)



Sampling Distribution of the Proportion (Different Sample Sizes)



Example 1: Effect of Sample Size on Standard Error of the Mean

Scenario

A company's delivery time has a population standard deviation of $\sigma = 20$ minutes. Compare the standard error for different sample sizes: $n = 10$, 30 and 100.

Solution

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{20}{\sqrt{n}}$$

Sample Size (n)	Standard Error (SE)
10	$20/\sqrt{10} = 6.32$
30	$20/\sqrt{30} = 3.65$
100	$20/\sqrt{100} = 2.00$

Interpretation

- When $n = 10$, the sample mean can vary by ± 6.32 minutes across different samples.
- When $n = 100$, the variation reduces to ± 2.00 minutes.
- A larger sample size gives a more precise estimate of the true mean delivery time.

Example 2: Effect of Sample Size on Standard Error of the Proportion

Scenario

A website has a true conversion rate of 8% ($p = 0.08$). Calculate the standard error for different sample sizes: $n = 50$, 200 and 400.

Solution

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.08(1-0.08)}{n}} = \sqrt{\frac{0.0736}{n}}$$

Sample Size (n)	Samplard Error (SE)
50	$\sqrt{(0.0736/50)} = 0.0384$
200	$\sqrt{(0.0736/200)} = 0.0192$
400	$\sqrt{(0.0736/400)} = 0.0136$

Interpretation

- When $n = 50$, the sample proportion can vary by about $\pm 3.84\%$.
- When $n = 400$, the variation reduces to $\pm 1.36\%$.
- A larger sample size gives a more precise estimate of the true conversion rate.

Business Use Cases

Market Research
Larger samples give more reliable estimates of customer preferences.

Quality Control
More samples reduce variation in defect rate estimates.

A/B Testing
Determine the right sample size for reliable conversion rates.

Surveys & Polls
Ensure accurate estimates of population proportions.

7 Point Estimate vs Interval Estimate

A point estimate gives a single value, while an interval estimate gives a range of values with a level of confidence.

Estimating population parameters using sample statistics.

What is a Point Estimate?

A **point estimate** is a single value calculated from a sample that is used to estimate a population parameter.

Point estimate of μ	$\rightarrow$ $\bar{x}$	(sample mean)
Point estimate of p	$\rightarrow$ $\hat{p}$	(sample proportion)

What is an Interval Estimate?

An interval estimate is a range of values calculated from a sample that is likely to contain the true population parameter with a certain level of confidence.

For mean (μ):	$\bar{x} \pm Z \times SE$
For proportion (p):	$\hat{p} \pm Z \times SE$

Key Differences

Feature	Point Estimate	Interval Estimate
Output	Single value	Range of values
Provides	One best guess	A range with confidence level
Includes margin of error	No	Yes
More reliable	Less reliable	More reliable
Example	$\bar{x} = 50$	$48.04 \leq \mu \leq 51.96$

Why Use Interval Estimate?

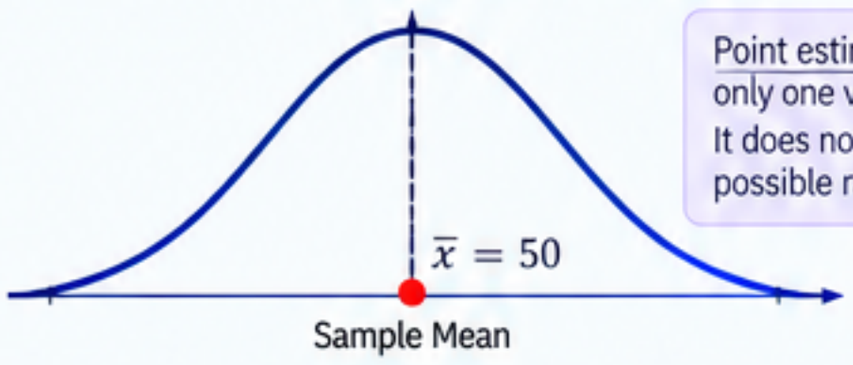
1. Provides a range of plausible values instead of a single value.
2. Takes into account sampling variability.
3. More reliable for decision making.
4. Used in business, analytics, and research.
5. Helps us quantify the uncertainty in the estimate.

Confidence Level and Z-Value

Confidence Level ($1 - \alpha$)	α	Z-Value
90%	0.10	1.645
95%	0.05	1.960
99%	0.01	2.576

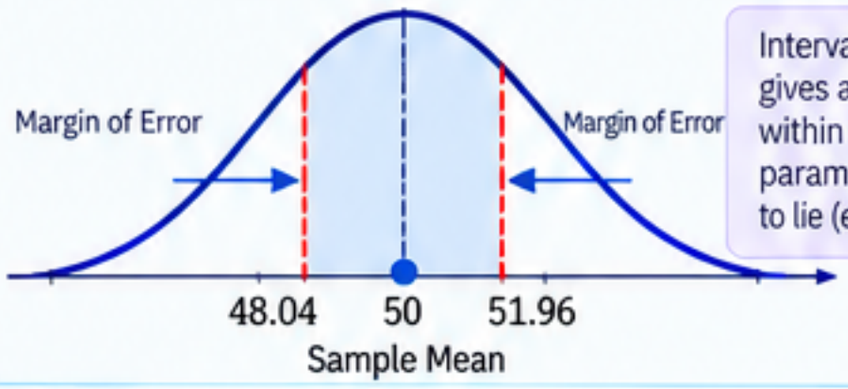
Visual Representation

Point Estimate (Single Value)



Point estimate gives only one value.
It does not show the possible range.

Interval Estimate (Range of Values)



Interval estimate gives a range of values within which the true parameter is likely to lie (e.g., 95% confidence).

Example 1: Point and Interval Estimate for Population Mean

Scenario

A company wants to estimate the average delivery time (μ). A random sample of $n = 100$ orders gives a sample mean of $\bar{x} = 30$ minutes. The population standard deviation is $\sigma = 10$ minutes. Find the point estimate and 95% confidence interval for the population mean.

Solution

- **Point Estimate:** $\bar{x} = 30$ minutes
- **Standard Error:** $SE = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{100}} = \frac{10}{10} = 1$
- **95% Confidence Interval:**

$$\bar{x} \pm Z \times SE = 30 \pm 1.96 \times 1$$
$$= 30 \pm 1.96 = (28.04, 31.96)$$

Interpretation

- Point estimate: 30 minutes (based on sample data).
- We are 95% confident that the true average delivery time lies between 28.04 and 31.96 minutes.
- This means if we repeatedly take samples, 95% of the intervals will contain the true mean.

Example 2: Point and Interval Estimate for Population Proportion

Scenario

A website has a sample of $n = 400$ visitors, and 48 made a purchase. Estimate the true conversion rate (p) using a point estimate and a 95% confidence interval.

Solution

- **Point Estimate:** $\hat{p} = \frac{48}{400} = 0.12\ (12\%)$
- **Standard Error:** $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.12(0.88)}{400}} = 0.0163$
- **95% Confidence Interval:**

$$\hat{p} \pm Z \times SE = 0.12 \pm 1.96 \times 0.0163$$
$$= 0.12 \pm 0.0319 = (0.0881, 0.1519)$$

Interpretation

- Point estimate: 0.12 (12%).
- We are 95% confident that the true conversion rate lies between 8.81% and 15.19%.
- This means if we take many samples of 400 visitors, 95% of the intervals will contain the true proportion.

8 Bias and Unbiased Estimators

Understanding bias in estimation and the concept of unbiased estimators.

An unbiased estimator gives the correct population value on average, while a biased estimator consistently overestimates or underestimates.

What is a Biased Estimator?

A **biased estimator** is a sample statistic whose expected value is **systematically different** from the true population parameter.

$$Bias(\hat{\theta}) = E(\hat{\theta}) - \theta$$

$E(\hat{\theta})$ = expected value of the estimator
θ = true population parameter

If $Bias(\hat{\theta}) \neq 0$, the estimator is biased.

What is an Unbiased Estimator?

An unbiased estimator is a sample statistic whose expected value is equal to the true population parameter.

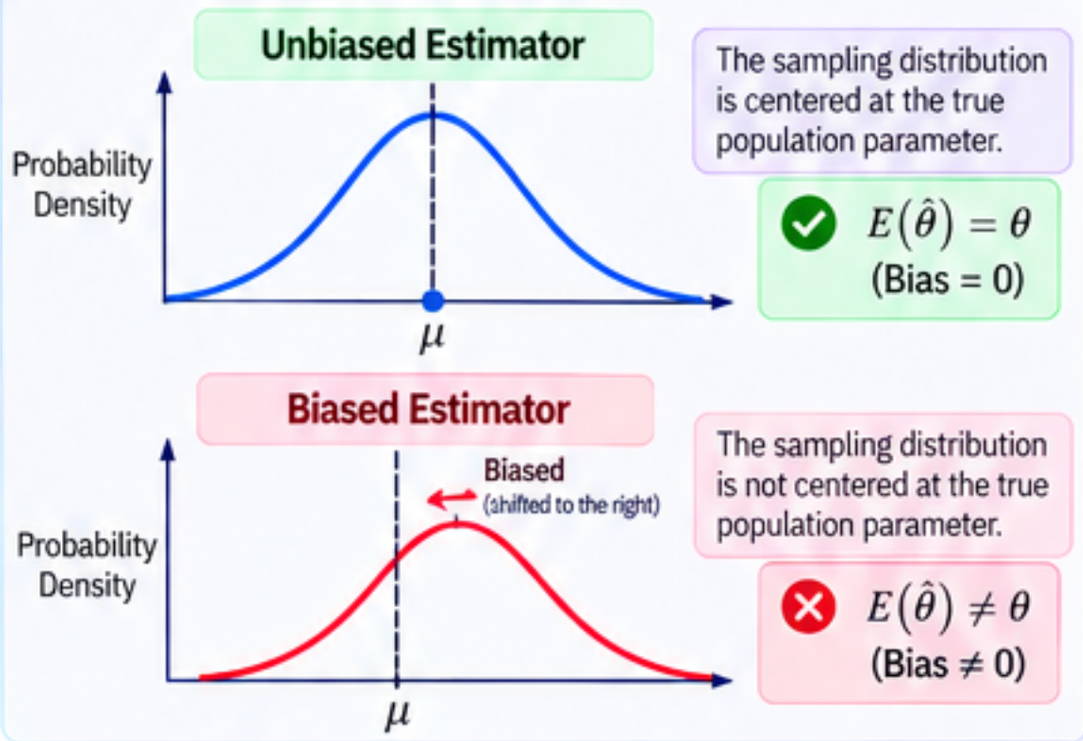
$$E(\hat{\theta}) = \theta \Rightarrow Bias(\hat{\theta}) = 0$$

- On average, it gives the correct population value.
- It may still vary from sample to sample (due to random variation), but it does not systematically overestimate or underestimate.

Examples of Biased and Unbiased Estimators

Parameter	Estimator	biased / Unbiased?	Reason
Population Mean (μ)	Sample Mean ($\bar{X}$)	Unbiased	$E(\bar{X}) = \mu$
Population Proportion (p)	Sample Proportion ($\hat{p}$)	Unbiased	$E(\hat{p}) = p$
Population Variance (σ^2)	$\frac{1}{n}\sum(X_i - \bar{X})^2$	Biased	Underestimates σ^2
Population Variance (σ^2)	$\frac{1}{n-1}\sum(X_i - \bar{X})^2$	Unbiased	$E(s^2) = \sigma^2$

Visual Comparison



Example 1: Biased Estimator for Variance (Using n instead of $n-1$)

Scenario

A population has values:
$[4, 6, 8, 10, 12]$
Population variance $(\sigma^2) = 10$
Take all possible samples of size $n = 2$ and calculate the sample variance using:

1. $\frac{1}{n}\sum(X_i - \bar{X})^2$ (Biased)
2. $\frac{1}{(n-1)}\sum(X_i - \bar{X})^2$ (Unbiased)

Show that the first one underestimates the true variance.

Solution

Sample Variance using n (Biased)

Sample	Mean ($\bar{X}$)	Variance $\left(\frac{1}{n}\right)\sum(X_i - \bar{X})^2$
[4, 6]	5	1
[4, 8]	6	4
[4, 10]	7	9
[4, 12]	8	16
[6, 8]	7	1
[6, 10]	8	4
[6, 12]	9	9
[8, 10]	9	1
[8, 12]	10	4
[10, 12]	11	1

Average of sample variances $= 5$

Sample Variance using $n-1$ (Unbiased)

Sample	Mean ($\bar{X}$)	Variance $\left(\frac{1}{n-1}\right)\sum(X_i - \bar{X})^2$
[4, 6]	5	2
[4, 8]	6	8
[4, 10]	7	18
[4, 12]	8	32
[6, 8]	7	2
[6, 10]	8	8
[6, 12]	9	18
[8, 10]	9	2
[8, 12]	10	8
[10, 12]	11	2

Average of sample variances $= 10$

- Using n underestimates the population variance (biased estimator).
- Using $n-1$ gives an unbiased estimator, which equals the true population variance (10).

Key Takeaways

1. A biased estimator systematically overestimates or underestimates the true parameter.
2. An unbiased estimator has an expected value equal to the true parameter.
3. The sample mean and sample proportion are unbiased estimators.
4. Using $n-1$ in the sample variance formula corrects the bias.
5. Unbiased estimators are preferred because they give correct results on average.

Business Use Cases

1. Estimating average sales, customer spending, or delivery times (unbiased estimator: sample mean).
2. Estimating conversion rate or defect rate (unbiased estimator: sample proportion).
3. Quality control: using correct variance estimator to avoid underestimating variability.
4. Market research: unbiased estimators help make reliable business decisions.
5. A/B testing: unbiased estimates ensure fair comparison.

Module 6 – Summary & Interview Questions

Sampling + CLT gives us a powerful tool to make reliable inferences about a population from a sample.

Key concepts, formulas, real-world applications and important interview questions from this module.

Key Takeaways from Module 6

1. The sampling distribution shows the distribution of a sample statistic (mean or proportion) across many samples.
2. According to the **Central Limit Theorem** (CLT), the sampling distribution of the sample mean is approximately normal for large sample sizes ($n \geq 30$), even if the population is not normal.
3. The mean of the sampling distribution of the mean equals the population mean (μ) and the standard error is $\sigma/\sqrt{n}$.
4. Conditions for CLT: random sampling, independence, sufficiently large sample size, and finite mean & standard deviation.
5. The standard error measures the variability of a sample statistic and decreases as the sample size increases.
6. A larger sample size leads to a more precise and stable estimate.
7. A point estimate gives a single value, while an interval estimate gives a range with a confidence level.
8. Biased estimators systematically overestimate or underestimate the true parameter, while unbiased estimators give the correct value on average.

Real-World Use Cases

Market Research
Estimate customer satisfaction, purchase intent, or brand preference using sample surveys.

Quality Control
Use sampling and CLT to estimate the average defect rate or product measurements.

A/B Testing
Compare conversion rates between two versions of a website using sample proportions and confidence intervals.

Business Forecasting
Estimate average sales, delivery time, or demand using sample data and quantify uncertainty.

Formulas at a Glance

Standard Error of Mean

$$SE(\overline{X}) = \frac{\sigma}{\sqrt{n}}$$

Standard Error of Proportion

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

Interval Estimate (Mean)

$$\overline{X} \pm Z \times SE$$

Interval Estimate (Proportion)

$$\hat{p} \pm Z \times SE$$

Bias of an Estimator

$$Bias\left(\hat{\theta}\right) = E\left(\hat{\theta}\right) - \theta$$

Unbiased Estimator

$$E\left(\hat{\theta}\right) = \theta$$

Important Interview Questions & Answers

1. What is a sampling distribution?

Answer: It is the probability distribution of a sample statistic (e.g., sample mean or sample proportion) across many samples of the same size drawn from a population.

2. State the Central Limit Theorem.

Answer: For a sufficiently large sample size ($n \geq 30$), the sampling distribution of the sample mean is approximately normal, regardless of the population distribution, with mean μ and standard error $\sigma/\sqrt{n}$.

3. What are the conditions for CLT?

Answer: Random sampling, independence of observations, sufficiently large sample size ($n \geq 30$), and finite population mean and standard deviation.

4. What is standard error? Why is it important?

Answer: It is the standard deviation of the sampling distribution of a sample statistic. It measures the variability of the estimate and helps in confidence intervals and hypothesis testing.

5. How does sample size affect standard error?

Answer: As sample size increases, standard error decreases at a rate of $1/\sqrt{n}$, leading to more precise estimates.

6. What is the difference between point estimate and interval estimate?

Answer: A point estimate gives a single value (e.g., $\overline{X}$ or $\hat{p}$), while an interval estimate gives a range of values that likely contains the true population parameter with a certain confidence level.

7. What is a 95% confidence interval?

Answer: It is a range of values calculated from sample data such that we are 95% confident the true population parameter lies within this range.

8. What is a biased estimator? Give an example.

Answer: A biased estimator has an expected value different from the true population parameter. Example: using $1/n$ instead of $1/(n-1)$ for sample variance, which underestimates the true variance.

9. What is an unbiased estimator?

Answer: An unbiased estimator has an expected value equal to the true population parameter. Example: sample mean $\overline{X}$ for population mean μ.

10. Why is CLT important in real life?

Answer: It allows us to make reliable conclusions about a population using sample data, even when the population distribution is unknown or not normal. It forms the basis for confidence intervals and hypothesis testing.



Module 6 Completed!

You've now mastered Sampling Distribution, Central Limit Theorem and Estimation.

"Small Samples. Big Insights."

Keep Practicing | Keep Learning | Keep Growing

See you in the next module!

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